A-A-54444
19 July 1991

# COMMERCIAL ITEM DESCRIPTION

## VENTILATOR, VOLUME, PORTABLE

The General Services Administration has authorized the use of this Commercial Item Description

This Commercial Item Description covers a self-contained, portable volume ventilator which operates on 11-30 volts DC and also on 110/230 volts AC, 50-400 Hz when used with the separate AC power supply furnished.

Salient Characteristics:

The portable volume ventilator containing time cycling, pressure limiting, constant flow, and microprocessor (CPU) control shall be suitable for use in hospitals, ambulances, fields, and in military aeromedical aircraft. It shall be suitable for providing controlled ventilations to the non-breathing patient and for assisting ventilations to the spontaneously breathing patient.

The ventilator shall be a self-contained apparatus except for a compressed gas source and shall contain at least the following: a control/display panel; internal battery; battery recharge circuit; EMI/RFI suppression circuitry; valve to select flow of gas; alarms; and chart depicting tidal volume by gas flow/inspiratory time. The apparatus with above mentioned items shall be self-contained and have its own separate carrying case with condensed operating instructions marked directly on the case or on a label attached to the case. These instructions are in addition to the specified Operating and Maintenance Manuals. The case shall be a semi-rigid plastic or corrosion-resistant metal material with integral carrying handle, and shall have a hinged, clear door to protect the operator controls of the control/display panel from inadvertent contact and provide splash protection.

AMSC N/A FSC 6530

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The internal battery shall be a rechargeable 12 volt battery. Two batteries wired to provide 12 volts is acceptable.

The ventilator battery shall be rechargeable from any of the following inputs: 11 to 30 volts DC, positive or negative ground; and nominal 110/230 volts AC, 50-400 Hz, single phase. A totally depleted internal battery shall be capable of being fully recharged within 24 hours from any of the voltages cited.

The ventilator apparatus shall include a patient valve with a removable PEEP valve (attached), all tubing and connectors required for complete ventilator system operation, a separate AC power supply with power cord assembly, and operating/maintenance manuals. All of the above, including the case containing the ventilator, shall be contained in a carry-all, semi-rigid carrying case.

The apparatus shall have a separate AC power supply. The AC power supply shall contain an Input Voltage selector switch to allow selection of either 110-125 volts, 50-400 Hz, single phase service or nominal 220-250 volts AC, 50-400 Hz, single phase service. This supply shall conform with the electromagnetic interference/radio frequency interference requirements. It shall also contain an appropriately strain-relieved line cord, at least 6 feet long, consisting of SJT-type wire with a minimum rating of 18/3 AWG, and terminating in a 3 conductor male hospital-grade plug. The output cable of this supply shall be at least 4 feet long and terminate in a suitable, non-shorting type plug to interface with the ventilator apparatus. The power supply output voltage shall be within the range of acceptance of the ventilator apparatus described herein.

The ventilator shall be operational from the 12 volt internal battery and from external DC sources of 11-30 volts including, vehicle generating system rated 120 amperes at 14 volts; 120 amp-hour heavy duty 12 volt battery, and aircraft 28 volt generating system. It shall operate from these external voltages irrespective of ground polarity (positive or negative). Additionally, the ventilator apparatus shall be operational from external AC power sources of 110/230 volts, 50-400 Hz, single phase when connected to the separate AC power supply furnished. It shall also be operational during battery recharge.



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A vehicle power cord assembly shall be provided approximately 6 feet long, having a vehicle cigar lighter plug on one end and a ventilator plug on the other end.

An aircraft DC power cord assembly shall be provided approximately 6 feet long, having one free end (no plug) and a ventilator plug on the other end.

Alternately, the vehicle power cord and/or aircraft power cord assemblies may be permanently attached to the ventilator.

The ventilator apparatus shall automatically perform a complete functional self-check prior to the commencement of operation. It shall also provide a means to calibrate the ventilator apparatus to ambient barometric pressure.

The ventilator shall have a minimum operating temperature range from minus 4°F to plus 115°F, from all power sources cited including the internal rechargeable battery. The minimum long-term storage temperature range shall be from 50°F to 80°F.

The unit shall be provided with a self-contained patient valve. The patient valve shall be reusable, autoclavable, and transparent. It shall be fabricated from material(s) compatible-for-use in environments of 100 percent oxygen. The patient valve shall include at least: a gas inlet port; combination 15mm (female)/22mm (male) patient connection port; anti-asphyxiation/demand valve inlet port; exhalation port with removable PEEP (positive end expiratory pressure) valve (attached); and pressure transducer adapter. The patient valve shall be configured to allow the ventilator apparatus to detect spontaneous inhalation efforts, and allow the ventilator to trigger at predetermined negative pressure thresholds, in conjunction with synchronized assist and synchronized intermittent mandatory ventilation modes of operation.

The apparatus shall also include a PEEP monitoring circuit, which when activated, enables the user to automatically monitor PEEP performance or manually program the apparatus for a specific PEEP value.



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The ventilator shall be provided with connecting tubing (with connectors) for attachment between the output connector of ventilator apparatus and gas inlet of patient valve; between the transducer input of ventilator apparatus and pressure transducer port of patient valve; and between gas input connector of ventilator apparatus and gas source of 50 psi.

The unit shall have an input to accept 100 percent oxygen, medical-grade air, or blended air/oxygen mixtures from an external blender with a nominal rating of 50 psi.

The ventilator shall have a gas-efficient patient circuit and not require purging of exhaled gas accumulations.

The ventilator shall have a flow rate variable from 0 to at least 100 liters per minute (LPM).

The unit shall have a ventilator rate adjustable from at least 1 to 150 ventilations per minute with a minimum resolution of 1 ventilation per minute (plus or minus 1 digit on the display panel).

Inspiratory times shall be adjustable from at least 0.1 to 3.0 seconds (as limited by the ventilator rate setting) with a minimum resolution of 0.1 seconds (plus or minus 1 digit on the display panel).

Inspiratory pressure (pressure limiting) shall be monitored and limited to a default maximum of 100 $cmH_2O$. It shall be adjustable over the entire high pressure alarm setting range. The ventilator apparatus shall provide a means by which to vent excess pressures away from the patient without disruption of the inspiratory time cycle.

The apparatus shall include a redundant means by which to inhibit gas flow to the patient in the event of CPU failure or CPU detected malfunction. During such occurrences, the ventilator apparatus shall cause the patient valve to default into a non-inspiration condition.

During inverse I:E ratio alarm conditions, the ventilator apparatus shall cause the patient valve to default into a non-inspiration condition.



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A sigh circuit shall be included, which when activated, shall provide a ventilator breath equivalent to 150 percent of the inspiratory time setting (truncated to a maximum of 3 seconds) once every 7 minutes or 100 breaths, whichever occurs first.

The synchronized assist and synchronized intermittent mandatory ventilation modes shall include an adjustable trigger level sensitivity to activate ventilator controlled breaths. The trigger level sensitivity range shall be adjustable to at least minus 8 $cmH_2O$. Trigger level sensitivities shall not be absolute values but values relative to positive end expiratory pressure (PEEP).

The ventilator shall have a manual trigger circuit. When its activating switch is depressed, the manual trigger shall permit a continuous flow of gas (at the flow control setting) until released. The manual trigger shall be operable during normal ventilator operation and in the event of microprocessor (CPU) failure. When the manual trigger is released the ventilator shall resynchronize for automatic operation by initiating one exhalation period before resuming timed breaths.

The unit shall have controls and indicators functionally grouped to facilitate operator usage.

A digital bar graph shall be included consisting of at least 20 distinct bars. The bar graph shall provide a means by which to continuously display airway pressure. The range of this graph shall be at least minus 10 to plus 100 $cmH_2O$.

The ventilator shall have a digital readout, consisting of at least 3 digits. This readout will furnish numeric data pertinent to airway conditions, control settings, alarms and calibration. The digital readout shall become activated whenever airway pressure, peak airway pressure, mean airway pressure and PEEP pressure display switches are depressed. It shall activate whenever the ventilator rate, inspiratory time, low airway pressure alarm and high airway pressure alarm control settings are adjusted or their respective display switches are depressed. The digital readout shall also activate at "power up" displaying the current settings for ventilator rate, inspiratory time, low and high pressure alarms. It shall activate to warn of CPU failure, an operator attempt to set inverse I:E ratio (inspiratory time greater than expiratory time), and detection of incorrect



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pressure transducer calibration. The digital readout shall be designed to activate at any time and incorporate a battery power saver which shall self-blank after approximately 3-seconds.

All controls, having a capability to activate the digital display shall have visual indicators which indicate the control function or airway condition being displayed. The indicator shall remain illuminated until the digital display self-blanks.

The unit shall have a visual indicator which illuminates when sigh control is activated.

A visual indicator which illuminates when the synchronized assist or synchronized intermittent mandatory ventilation modes are activated shall be provided.

Additional indicators which denote the presence of operating power and/or connection to live external power sources shall be provided.

The ventilator shall contain a pleurality of alarm circuits. Alarms shall provide both audible and visual indications. For operator convenience and safety, certain alarms shall incorporate audible muting, automatic reset, automatic reactivation and mute overrides as described. Alarms such as CPU failure, inverse I:E and pressure transducer calibration shall not be mutable.

A new alarm occurrence shall override any pre-existing mute condition(s). Each of these alarms shall automatically reset when the alarm condition no longer exists and reactivate should the condition present itself again.

The unit shall have as a minimum, alarms which monitor the internal battery, external power, high and low airway pressure conditions, disconnect, apnea, PEEP, CPU failure, inverse I:E ratio and pressure transducer calibration.

The apparatus shall have a battery alarm that will activate when conditions indicative of low battery charge or shorted battery are sensed. The ventilator apparatus shall be operable from external power regardless of battery charge status. Battery alarm conditions shall be capable of being audibly muted for 5-minute periods.



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The ventilator shall have an external power alarm that shall activate when a low external voltage condition is sensed. It shall also activate when no external voltage is sensed and a power source plug is connected to the ventilator apparatus. When the external power alarm activates, the ventilator apparatus shall immediately begin operation from its internal batteries. The external power alarm shall be mutable, and remain muted, until the internal battery alarm activates. Thereafter, the mute period shall last for 5-minute periods.

The ventilator shall have high pressure alarm settings that are adjustable from 15 to 100 $cmH_2O$ (100 shall be equivalent to "OFF") with a minimum resolution of 1 $cmH_2O$ (plus or minus 1 digit on the display panel).

The unit shall have a high pressure alarm that shall operate in conjunction with the high pressure alarm control setting. It shall be mutable for 30 second periods.

The apparatus shall have low pressure alarm settings that are adjustable from "OFF" to 50 $cmH_2O$ with a minimum resolution of $1cm_2H_2O$ (plus or minus 1 digit on the display panel).

A low pressure alarm that shall operate in conjunction with the low pressure alarm control setting shall be provided. It shall also detect the absence of breathing gas(es) and activate when the pressure transducer fails to detect a pressure rise in the patient circuit during ventilator delivered breaths. It shall be mutable for 30 second periods.

The ventilator shall have a disconnect alarm that shall operate when the pressure transducer fails to detect a pressure rise in the patient circuit during ventilator delivered breaths. The disconnect alarm shall be mutable for 30-second periods.

The PEEP alarm shall activate when the CPU fails to ascertain PEEP pressure. The PEEP alarm shall be mutable for 30-second periods.

An apnea alarm that shall activate when the patient fails to initiate a spontaneous breath during assisted or SIMV operating modes shall be provided. The apnea alarm circuit shall test to determine whether the condition is apnea or patient circuit disconnect/low pressure before sounding the appropriate alarm. When the apnea alarm is activated, the



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ventilator apparatus shall include a means by which to automatically activate assist-control breathing at the ventilator rate/inspiratory time setting, or 12 breaths per minute/inspiratory time setting, whichever is greater. The apnea alarm shall be cancelable, not mutable.

The unit shall have a CPU failure alarm that shall activate whenever the CPU fails to successfully complete a functional self-check when the ventilator apparatus is placed in an operating mode or detects an operational fault during the course of operation. The CPU failure alarm shall not be mutable or cancelable.

An inverse I:E ratio alarm that shall activate whenever inspiratory time exceeds expiratory time shall be provided. This alarm shall only be canceled by the operator lessening the inspiratory time control setting or lessening the rate control setting. The inverse I:E alarm shall not be mutable.

The ventilator shall have a pressure transducer calibration alarm that shall activate when the CPU detects a "non-zero" pressure transducer reading when exposed to an open atmosphere. The pressure transducer calibration alarm shall not be mutable. The alarm shall only be canceled by recalibrating the pressure transducer.

The apparatus shall have a battery operating time of at least 11 hours of continuous operation without diminishment of operating capability. Battery shall not deplete more than approximately 1.5 percent per month or 18 percent per year if stored at 70°F.

All circuitry within the ventilator and separate AC power supply shall be designed to conform within the electromagnetic interference/radio frequency interference requirements.

Dimensions of the self-contained ventilator apparatus shall not exceed 10 inches by 5.0 inches by 12 inches.

The self-contained ventilator apparatus, less all other components, shall not weigh more than 12 pounds.

Marking case containers. If the unit requires restricted conditions of storage, then the exterior markings of the over packaging, the carry-all carrying case, and the self-contained ventilator separate container shall indicate the required storage temperature and humidity.



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Workmanship. The portable volume ventilator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One ventilator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of 1. The unit of product for examination purposes shall be one ventilator complete unit.

Sampling for tests. Sampling for tests (dimensional), shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of 1. The unit of product for test purposes shall be one ventilator complete unit.



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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.



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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6530-01-292-1049

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - OSS
USPHS
FDA-MPQAS

Project Number: 6530-2150

Location: VENTMC12.DOC



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2 The submitter of this form must complete blocks 4, 5, 6, and 7.

3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I. RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER: A-A-54444 | 2. DOCUMENT DATE (YYMMDD): 12 July 1991 |
|---|---|---|

3. DOCUMENT TITLE: VENTILATOR, VOLUME, PORTABLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>DEFENSE PERSONNEL SUPPORT CENTER<br>ATTN: DPSC-MSE | b. TELEPHONE (Include Area Code)<br>(1) Commercial: AC 215-737-2870<br>(2) AUTOVON: AV 444-2870 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>2800 SO. 20TH STREET<br>PHILADELPHIA, PA 19101-8419 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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